

DeepSense 6G V2V Scenario 36: Fully Annotated Dataset Guide

Comprehensive Feature Specification, Coordinate Reference Systems, Codebook Geometry, and Drive-Block Splitting

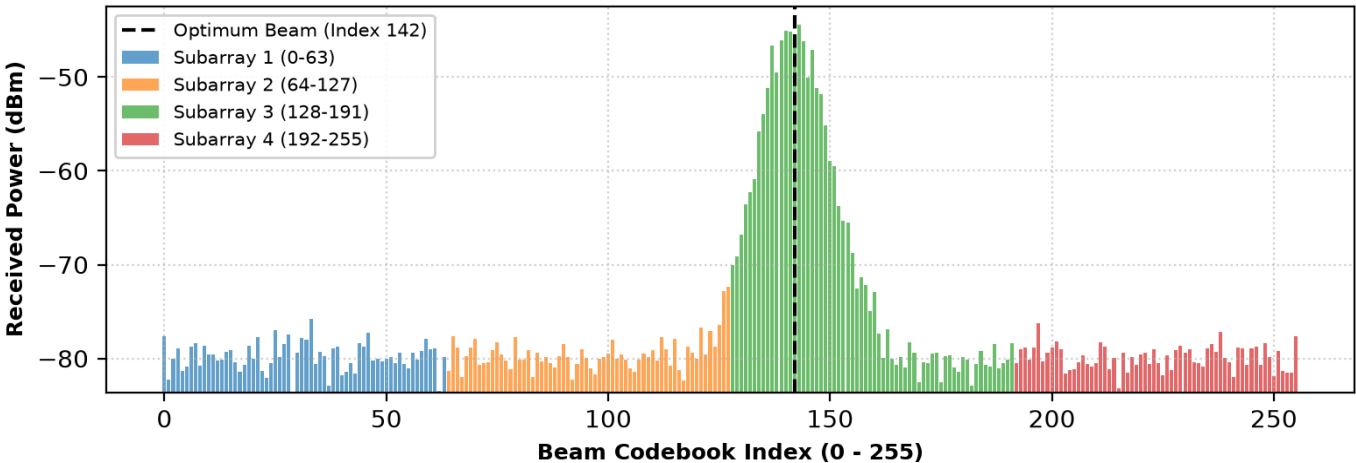
1. Dataset Overview & Vehicle Setup

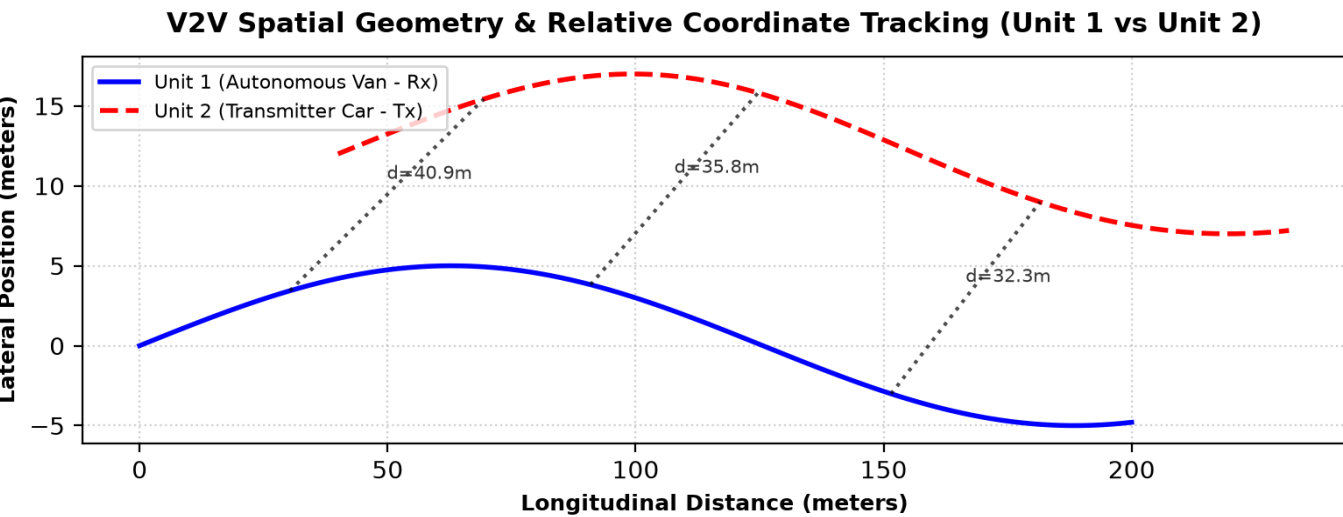
The **DeepSense 6G Scenario 36** dataset is a large-scale real-world multi-modal vehicle-to-vehicle (V2V) dataset collected in downtown and suburban driving conditions. It captures real-world interactions between two dynamic vehicles operating at 60 GHz millimeter-wave (mmWave) frequencies:

Vehicle	Role	Hardware & Sensors	Sampling Rate
Unit 1 (Ford Transit)	Receiver (Rx)	4x 64-element 60 GHz Phased Arrays (256 beams), ZED2 RGB Camera, RTK-GPS	10 Hz
Unit 2 (Lincoln MKZ)	Transmitter (Tx)	60 GHz Omni-directional Transmitter Horn Antenna, RTK-GPS	10 Hz

2. Spatial Kinematics & 256-Beam Codebook Structure

DeepSense 6G 256-Element Millimeter-Wave Beam Spectrum Across 4 Subarrays





3. Complete Feature Schema & Annotations

Feature Category	Variable Name	Data Type	Physical Unit	Description & Interpretation
Spatial Kinematics	rel_lat	float32	degrees	Relative latitude offset (Unit 2 Lat - Unit 1 Lat)
Spatial Kinematics	rel_lon	float32	degrees	Relative longitude offset (Unit 2 Lon - Unit 1 Lon)
Spatial Kinematics	distance	float32	meters (m)	Euclidean distance between Unit 1 and Unit 2 (d in meters)
Spatial Kinematics	bearing	float32	radians	Line-of-sight azimuth angle $\theta = \arctan2(\text{rel_lon}, \text{rel_lat})$
Spatial Kinematics	sin / cos_bearing	float32	unitless	Trigonometric projections eliminating 0-2pi boundary discontinuity
Spatial Kinematics	bearing_rate	float32	rad/s	First-order temporal derivative (angular velocity of LOS beam)
Spatial Kinematics	distance_rate	float32	m/s	First-order range rate (relative radial closing speed)
Vehicle Dynamics	unit1_speed / hdg	float32	m/s, deg	Unit 1 instantaneous driving speed and compass heading
Vehicle Dynamics	unit2_speed / hdg	float32	m/s, deg	Unit 2 instantaneous driving speed and compass heading
Visual Modality	unit1_rgb5 / rgb6	RGB Image	96x96x3	Forward-facing ZED2 wide-angle camera visual scene stream
RF Power Ground Truth	pwr_0 ... pwr_255	float32	linear mW	Received power values across all 256 candidate beams
Target Variable	unit1_overall-beam	int64	0 ... 255	Optimal ground-truth beam index maximizing received power

4. Drive-Based Split & Zero-Leakage Architecture

Standard chronological data splitting introduces severe time-series leakage because overlapping temporal windows (e.g., $T_{\text{gps}}=30$ steps) share 29 time steps between adjacent samples. Our pipeline groups samples strictly by continuous trajectory run IDs (seq_index, 124 drives total), ensuring zero temporal or spatial leakage across splits:

Dataset Split	Drive Runs	Sample Count	Percentage	Role in Pipeline
Train Set	86 continuous drives	17,199 samples	70.0%	Neural network backpropagation & parameter optimization
Calibration Set	18 continuous drives	3,600 samples	15.0%	Static Conformal & Conformal PID quantile calibration
Test Set	20 continuous drives	4,000 samples	15.0%	Unseen benchmark evaluation & APL / Power Ratio calculation

5. Conformal PID Feedback Control & Latency Reduction

Rather than probing all 256 candidate beams sequentially (which induces high latency and packet drops in high-speed V2V mobility), our **Conformal PID Feedback Controller** dynamically bounds beam search space with mathematical coverage guarantees:

Control Method	Target Miss Rate (α)	Empirical Test Miss Rate	Avg Candidate Beams	Beam Sweep Reduction
Full Sweep Baseline	0.0% (exhaustive)	0.0%	256.0 beams	0.0% (Baseline)
Static Conformal ($q=8.0\text{dB}$)	10.0%	17.8%	140.7 beams	45.1% reduction
Online Adaptive (ACI)	10.0%	31.7%	18.7 beams	92.7% reduction (High Misses)
Conformal PID ($P=0.04$, $I=0.0008$, $D=0.008$)	10.0%	7.5% (Safe $\leq 10\%$)	169.8 beams	33.7% reduction